

CS105 (DIC on Discrete Structures)

Exercise problem set 12

- Attempt *all* questions.
 - Apart from things proved in lecture, you cannot assume anything as “obvious”. Either quote previously proved results or provide clear justification for each statement.
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Part 1

1. Determine the maximum number of edges in a bipartite subgraph of the following graphs:
 - P_n : the path graph on n vertices,
 - C_n : the cycle graph on n vertices,
 - K_n : the complete graph on n vertices.
2. Determine the minimum size of a maximal matching in the cycle graph C_n , where C_n denotes a cycle on n vertices.
3. Let S be the set of vertices saturated by a matching M in a graph G . Prove that some maximum matching also saturates all of S . Must the statement be true for every maximum matching?

Part 2

4. For $k \geq 2$, prove that a k -regular bipartite graph has no cut-edge, where a *cut-edge* (or *bridge*) is an edge whose removal increases the number of connected components of the graph.
5. Let G be an n -vertex simple graph, where $n \geq 2$. Determine the maximum possible number of edges in G under each of the following conditions:
 - (a) G has an independent set of size α .
 - (b) G has exactly k components.

- (c) G is disconnected.
6. Prove or disprove: If a (simple) graph has no cycles then it has at most one perfect matching.
 7. Prove or disprove: every maximal matching in a graph G has at least half the size of a maximum matching.